JavaScript Loops — Cheat Sheet & Guide

JavaScript mein loops ka istemal code ko repeat karne aur repetitive tasks ko automate karne ke liye kiya jata hai.

1. For Loop — Standard Loop

Kaam kya hota hai?
Code ko ek fixed number of times repeat karta hai.

Kyu use hota hai?
Jab aapko pehle se pata ho ki loop kitni baar chalana hai.

```
let a = 1;
for (let i = 0; i < 10; i++) {
    console.log(a + i); // 1 se 10 tak print karega
}
```

2. For-In Loop — Objects Ke Liye

Kaam kya hota hai?
Kisi Object ki saari keys (properties) par ek-ek karke jaata hai.

Kyu use hota hai?
Object ki keys ya unki values ko read/iterate karne ke liye.

```
let obj = {
    name: "harry",
    role: "programmer",
    company: "CodeWithHarry"
};

for (const key in obj) {
    console.log(key); // Prints: name, role, company
}
```

3. For-Of Loop — Arrays/Strings Ke Liye

Kaam kya hota hai?
Iterable data (jaise String ya Array) ke har ek element par jaata hai.

Kyu use hota hai?
String ke characters ya Array ke items ko direct access karne ke liye.

```
let iterator = "harry";

for (const c of iterator) {
    console.log(c); // Prints: h, a, r, r, y
}
```

4. While Loop Condition-Based

Kaam kya hota hai?
Pehle condition check karta hai, fir andar ka code chalata hai.

Kyu use hota hai?
Jab tak condition TRUE hai tab tak code chalane ke liye.

```
let i = 0;
while (i < 6) {
    console.log(i); // 0 se 5 tak print karega
    i++;
}
```

5. Do-While Loop At Least Once

Kaam kya hota hai?
Pehle ek baar code chala deta hai, uske BAAD condition check karta hai.

Kyu use hota hai?
Jab code ko KAM SE KAM EK BAAR chalana zaroori ho, chahe condition FALSE hi kyu na ho.

```
let j = 0;
do {
    console.log(j); // Minimum 1 baar zaroor chalega
    j++;
} while (j < 6);
```

Quick Summary / Ek Nazar Mein

Loop Type	Kiske Liye Best Hai?	Key Feature
For Loop	Numbers / Fixed iterations	Pata ho kitni baar chalana hai
For-In	Objects (Key-Value)	Keys par loop karta hai
For-Of	Strings & Arrays	Direct values/elements deta hai
While	Unknown repetitions	Pehle condition check karta hai
Do-While	Guaranteed execution	Kam se kam 1 baar zaroor chalega